

VGGT for 3D Reconstruction and Beyond: A Survey of Geometric State Strengthening and Its Applications

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Visual Geometry Grounded Transformer (VGGT) jointly predicts cameras, point maps, depth maps, and tracks from multiple images in a feed-forward manner. *It provides multi-view geometry latent features and structured geometric outputs (i.e., cameras, point maps, depth maps, and tracks).* Denoted by $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$, these two components together form the geometric state examined in this survey. From this geometric-state perspective, we organize the literature into two directions: state-strengthening methods improve how this state is formed from diverse inputs, computed efficiently, made robust, retained over streams, or extended to dynamic scenes, whereas state-reuse methods use geometric state for novel-view synthesis, simultaneous localization and mapping (SLAM), semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Beyond the method taxonomy, we organize the datasets, metrics, and evaluation targets used to assess geometric-state strengthening and downstream reuse. Finally, we point out open challenges for future research. Our open-source resources are available at https://richardchen225.github.io/vggt_survey/.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Computer vision**; **Reconstruction**.

Additional Key Words and Phrases: 3D vision, geometric foundation models, feed-forward reconstruction, geometric state, point maps, visual geometry grounded transformer

1 Introduction

3D reconstruction is a long-standing problem in computer vision that provides the geometric foundation for downstream tasks such as novel-view synthesis [2, 75], localization and mapping [109, 110], semantic 3D scene understanding [131, 189], geometry-aware world models [129, 153], and embodied action and planning [1, 194]. Over the past several years, 3D reconstruction has progressed from scene-specific optimization with neural representations such as NeRF [115] and 3D Gaussian Splatting [80] toward feed-forward reconstruction represented by DUST3R-style [175] and VGGT-style [167] models.

Feed-forward reconstruction amortizes the reconstruction process across scenes, allowing a model to predict camera or scene geometry directly from sparse, unposed, or partially constrained observations rather than fitting a separate scene-specific model. A recent influential line of feed-forward reconstruction is represented by point-map-based models in the DUST3R family, including

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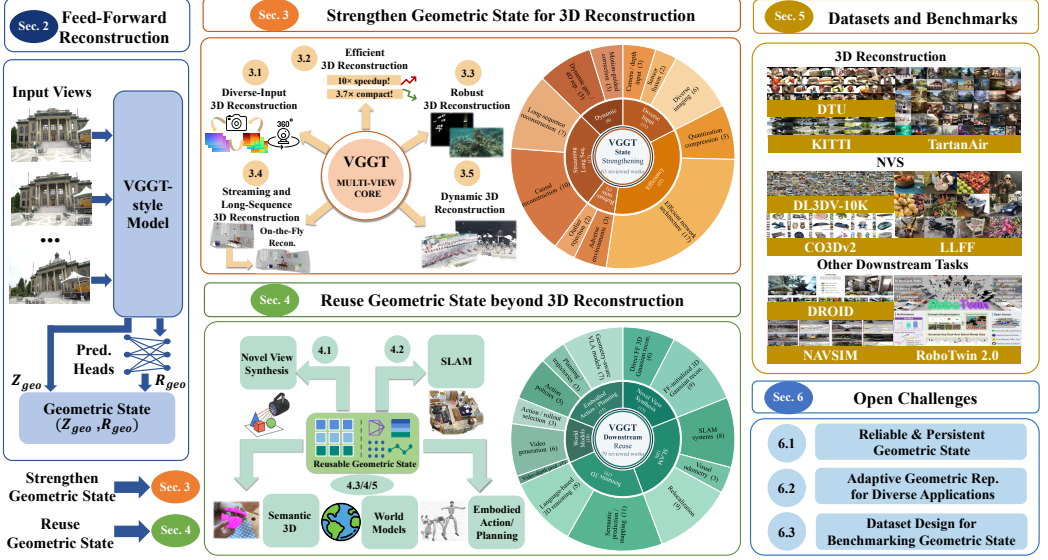


Fig. 1. Overview of the survey organization. Feed-forward reconstruction produces a geometric state containing multi-view geometry latent features Z_{geo} and structured geometric outputs R_{geo} . State-strengthening methods improve how this state is formed, computed, made robust, retained over streams, or extended to dynamic scenes. Downstream methods reuse geometric state for novel-view synthesis, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Datasets and evaluation targets assess the reviewed methods, and the open challenges concern reliable and persistent state, adaptive representations, and dataset design for benchmarking geometric state. Numbers in parentheses indicate the number of reviewed works in each category or subcategory.

DUST3R [175], MAST3R [89], Fast3R [188], MUST3R [12], and CUT3R [172], which made pixel-aligned 3D structure a practical interface between images and scene geometry. DUST3R [175] uses an image pair as its basic inference unit, so a larger image collection must still be assembled through global alignment. Building on the direct image-to-point-map formulation of DUST3R [175], the field has progressed to VGGT-style [167] models that jointly infer structured geometric outputs and learn reusable multi-view geometry latent features.

Although existing surveys have reviewed neural scene representations [182], 3D Gaussian Splatting [18], and feed-forward 3D reconstruction from DUST3R to VGGT [204, 208], the rapidly expanding VGGT [167] lineage—including numerous architectural extensions and downstream adaptations—has received limited systematic analysis. A dedicated survey is therefore needed to clarify how these works relate to one another, assess the current state of the field, and identify a roadmap for future research.

To organize this fragmented literature, we review VGGT-style [167] models through their geometric state, comprising multi-view geometry latent features Z_{geo} and structured geometric outputs R_{geo} . We divide existing studies into two branches: state strengthening and state reuse. As illustrated in Fig. 1, state strengthening improves how the geometric state is constructed and maintained, whereas state reuse adapts or consumes the state for downstream applications.

State strengthening. The first branch improves how the geometric state is formed from diverse inputs, computed efficiently, made robust, retained across long sequences, or extended to dynamic

scenes. Accordingly, Section 3 organizes these studies into diverse-input, efficient, robust, streaming and long-sequence, and dynamic 3D reconstruction.

State reuse. The second branch uses the structured geometric outputs, multi-view geometry latent features, or both components of the geometric state to support downstream applications beyond feed-forward reconstruction. Section 4 examines how 5 downstream application groups—novel-view synthesis, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning—use these components. Application models may use the state produced by VGGT [167] or by a strengthened variant, depending on the information required by the target application.

Datasets and benchmarks. To support meaningful comparison across both branches, Section 5 organizes the datasets, metrics, and evaluation targets used to assess geometric-state strengthening and downstream reuse. This analysis reveals substantial differences in evaluation coverage: static reconstruction and rendering have relatively mature resources, whereas long-stream, dynamic, semantic, embodied, and world-model settings depend more heavily on specialized datasets.

Open challenges. Based on the reviewed methods and evaluation evidence, Section 6 identifies 3 unresolved problems: building reliable and persistent geometric state, designing adaptive geometric representations for diverse applications, and designing datasets and benchmarks that determine whether application gains result from changes in or reuse of geometric state. Together, these challenges follow the survey’s progression from strengthening the geometric state, to reusing it in downstream applications, and finally to evaluating what the state contributes.

Contribution. To summarize, this survey provides the first systematic review centered on the VGGT [167] lineage, introduces a geometric-state perspective that unifies state strengthening and downstream reuse, consolidates the associated datasets and evaluation targets, and identifies key directions for future research. Our goal is to help researchers understand the rapidly evolving VGGT [167] landscape, relate existing methods within a common framework, and identify promising opportunities for further development in this growing field.

2 Feed-Forward 3D Reconstruction: Foundations and VGGT Model

This section first reviews the evolution from scene-specific optimization to feed-forward 3D reconstruction. It then introduces VGGT [167] and defines its geometric state. Finally, we discuss the model limitations that motivate Sections 3 and 4.

2.1 Evolution toward Feed-Forward 3D Reconstruction

Classical 3D reconstruction solves each scene through an optimization process. Structure-from-Motion (SfM) [139] estimates camera poses and sparse structure from image correspondences, after which Multi-View Stereo (MVS) [140] densifies the recovered geometry. NeRF [115] and 3D Gaussian Splatting (3DGS) [80] replace conventional geometric representations with neural radiance fields or optimizable Gaussian primitives, but their original formulations still fit scene-specific parameters from posed observations. Although these approaches differ in representation and rendering, geometry becomes available only after optimization has been performed for the target scene.

Feed-forward reconstruction moves reconstruction from per-scene optimization to amortized inference. After learning across many scenes, a model can predict camera or scene geometry directly from sparse, unposed, or partially constrained observations. The amortized formulation makes geometric structure an immediate network output rather than only the endpoint of a per-scene solver.

DUST3R [175] instantiated the feed-forward paradigm through point maps, assigning each image pixel a 3D coordinate in a common coordinate frame within an image pair. Its basic inference unit is

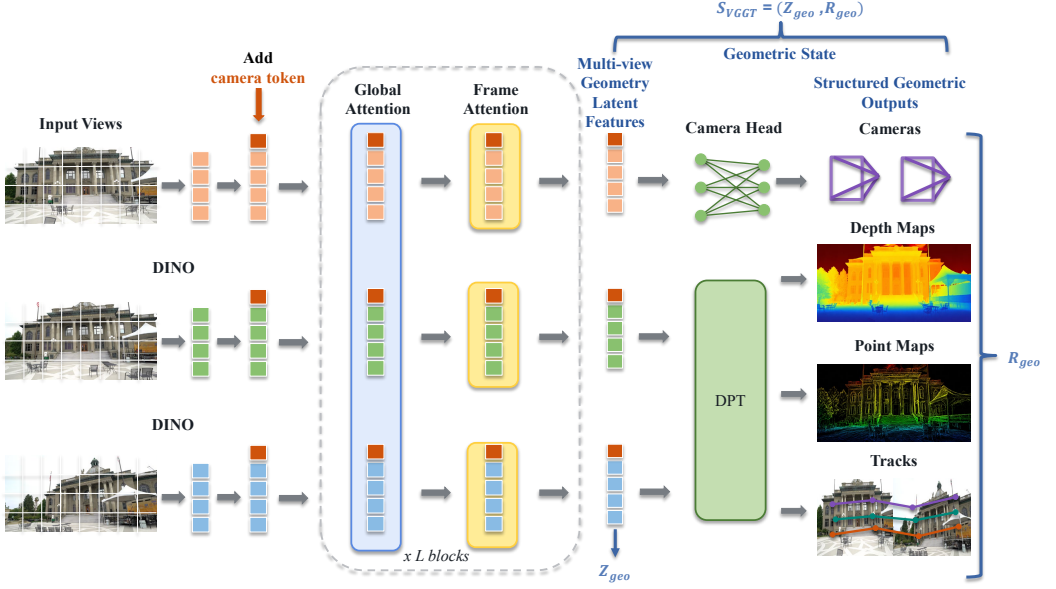


Fig. 2. VGGT [167] architecture. Input images are converted into tokens and augmented with camera and register tokens. Alternating frame-wise and global attention produces the multi-view geometry latent features Z_{geo} . Prediction heads decode cameras, depth maps, point maps, and tracks as the structured geometric outputs R_{geo} .

an image pair: the network predicts pixel-aligned point maps for both views, while reconstruction over a larger image collection requires predictions from multiple pairs to be assembled through a subsequent global alignment stage. Camera parameters can be recovered from the aligned geometry, but they are not jointly decoded with a complete multi-view scene representation in the same forward pass.

Point maps therefore establish a direct image-to-3D mapping, but DUST3R [175] still forms a coherent multi-view reconstruction by aligning pairwise predictions. This limitation motivates VGGT [167], which learns a set-level multi-view representation from which cameras, point maps, depth, and tracks can be decoded together. Its formulation and architecture are introduced next.

2.2 Introduction to VGGT

Given a set of input images, VGGT [167] aggregates multi-view evidence into a shared latent representation and predicts camera parameters, dense point maps, depth maps, and 3D point tracks in a single forward pass. This brings camera estimation, depth prediction, point-map reconstruction, and tracking into a single learned inference stage. Unlike pairwise point-map methods, it processes the image set jointly and exposes the resulting scene geometry through task-specific prediction heads.

VGGT [167] uses a multi-view transformer to aggregate information across the image set. Alternating frame-wise and global attention allows within-view structure and cross-view evidence to interact before camera, point-map, depth, and track heads make their predictions. Fig. 2 summarizes this architecture, from image tokenization and multi-view fusion to the 4 structured outputs.

More concretely, DINOv2 [118] patchifies each image into image tokens, to which VGGT [167] appends a camera token and register tokens for that view. The first view receives distinct learned

tokens so that it can serve as the reference frame. For an image set $\mathcal{I}_{\text{img}} = \{I_i\}_{i=1}^N$, the backbone can be summarized as

$$\begin{aligned} X_i^{(0)} &= [E_{\text{DINO}}(I_i), z_i^{\text{cam}}, Z_i^{\text{reg}}], \\ X^{(\ell+1)} &= \mathcal{A}_{\tau(\ell)}^{(\ell)}(X^{(\ell)}), \quad \tau(\ell) \in \{\text{frame}, \text{global}\}. \end{aligned} \quad (1)$$

Here I_i is the i th input image, N is the number of views, and E_{DINO} is the DINOv2 [118] image encoder that maps I_i to patch tokens. The learned token z_i^{cam} carries view-level camera information, while Z_i^{reg} denotes the bank of register tokens appended to view i ; $X_i^{(0)}$ is their concatenation with the patch tokens, and $X^{(\ell)} = \{X_i^{(\ell)}\}_{i=1}^N$ collects the tokens at transformer stage ℓ . The operator $\mathcal{A}_{\tau(\ell)}^{(\ell)}$ is the ℓ th attention block, and $\tau(\ell)$ specifies whether that block applies frame-wise or global attention. Frame-wise attention processes each $X_i^{(\ell)}$ separately, whereas global attention processes the tokens concatenated across views. The intermediate image tokens across these stages form the hierarchy later denoted by $\mathcal{Z}_{\text{geo}}$.

The prediction heads read different parts of this shared hierarchy. We denote the camera-head output for view i by $g_i = [q_i, t_i, f_i]$, where $q_i \in \mathbb{R}^4$ is a rotation quaternion, $t_i \in \mathbb{R}^3$ is translation, and $f_i = (f_i^h, f_i^v)$ contains the horizontal and vertical fields of view; the first camera defines the reference coordinate frame. A Dense Prediction Transformer (DPT) decoder [167] combines image tokens from multiple backbone stages to produce dense depth maps, point maps, tracking features, and confidence estimates for depth and points. The tracking module then consumes the dense tracking features together with a set of query points Q and predicts their cross-view locations and visibility. Thus tracks are query-conditioned outputs, whereas cameras, depth, and point maps are decoded directly for every input view.

The outputs are separate but architecturally coupled through end-to-end multi-task training, which can be schematically summarized as

$$\mathcal{L} = \mathcal{L}_{\text{camera}} + \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{pmap}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}}. \quad (2)$$

Camera parameters are supervised with a robust regression loss; depth and point maps use confidence-weighted value and gradient terms; and the tracking objective includes correspondence and visibility supervision. The prediction heads share multi-view features through the backbone, but joint training does not by itself enforce exact geometric agreement among their predictions.

Taken together, the architecture exposes two related forms of geometric information. Before decoding, the intermediate transformer stages retain multi-view geometry latent features shaped by the joint camera, point-map, depth, and tracking objectives. After decoding, the prediction heads provide structured geometric outputs, including cameras, point maps, depth maps, and query-conditioned tracks, that can be inspected or used by another application model. This distinction is central to the later literature: methods may modify the internal representation, operate on the decoded geometry, or combine both components in a downstream application. The next subsection formalizes these two parts as the geometric state used throughout the survey.

2.3 Introduction to Geometric State

To compare methods that access different parts of VGGT [167], we formalize the *geometric state* as an analytical abstraction for this survey. This abstraction separates the internal multi-view representation from the structured quantities decoded from that representation:

$$\mathcal{S}_{\text{VGGT}} = (\mathcal{Z}_{\text{geo}}, \mathcal{R}_{\text{geo}}), \quad (3)$$

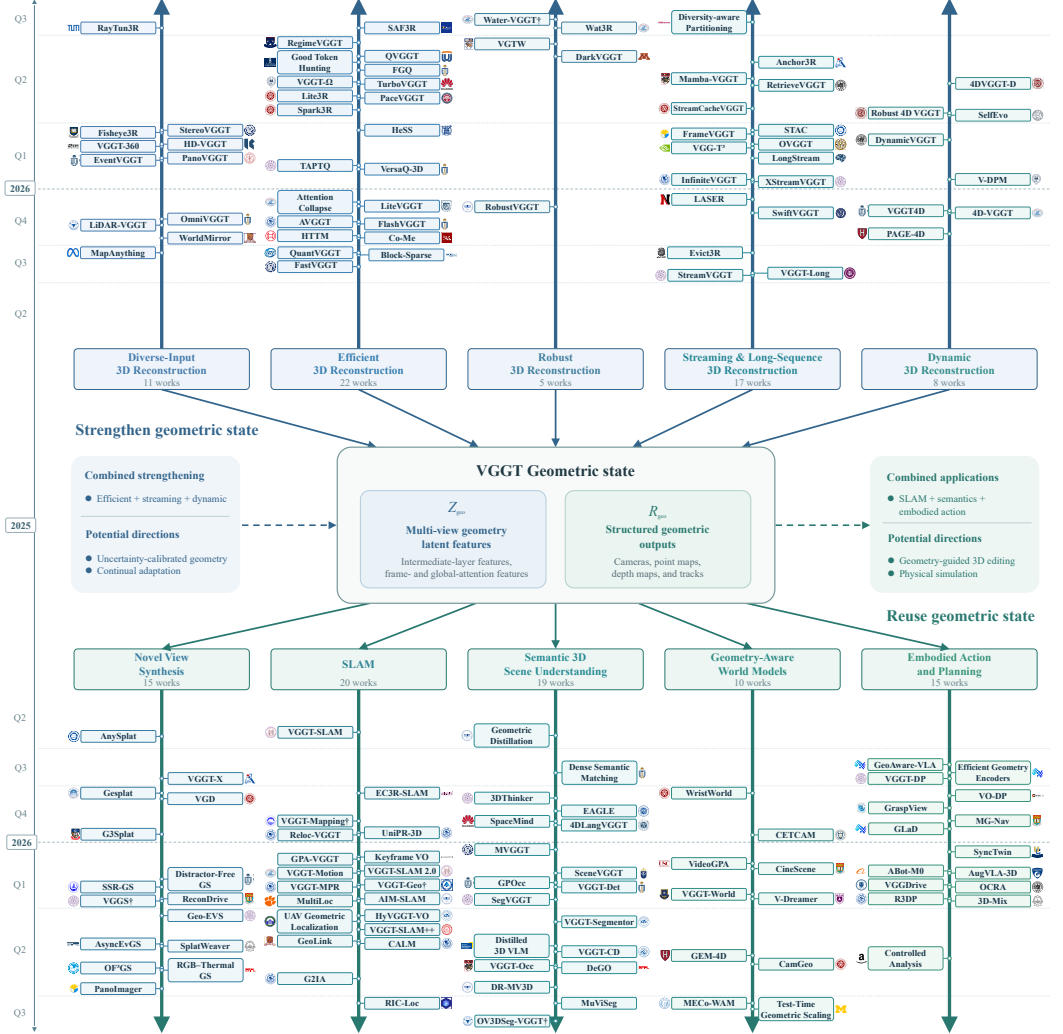


Fig. 3. Development of the VGGT [167] lineage around geometric state. The center contains multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$ and structured geometric outputs $\mathcal{R}_{\text{geo}}$. The 5 upper branches organize geometric-state strengthening methods, and the 5 lower branches organize downstream applications. Methods follow chronological order outward from the center, with years and quarters marked on the left. The side panels illustrate combinations of strengthening directions and applications, together with potential directions.

Equation (3) defines one geometric state with two components: $\mathcal{Z}_{\text{geo}}$ denotes multi-view geometry latent features, and $\mathcal{R}_{\text{geo}}$ contains structured geometric outputs. For VGGT [167], $\mathcal{S}_{\text{VGGT}}$ denotes the base geometric state. Later methods may strengthen how these components are formed, retained, or decoded.

The multi-view geometry latent features are collected from selected stages of the backbone:

$$\mathcal{Z}_{\text{geo}} = \left\{ \mathcal{Z}_{\text{geo}}^{(\ell)} \right\}_{\ell \in \mathcal{L}}, \quad (4)$$

where $\mathcal{L}$ denotes whichever backbone stages a particular method exposes or selects. Thus $\mathcal{Z}_{\text{geo}}$ denotes the selected feature levels rather than a fixed tensor shared by every method. These features are learned through multi-view aggregation and joint camera, point-map, depth, and track supervision; correspondence-like and epipolar structure observed in the attention layers provides additional evidence that these features encode more than appearance alone [8].

For N views and a set of track queries Q , $\mathcal{R}_{\text{geo}}$ collects the structured geometric outputs used by the later taxonomy:

$$\mathcal{R}_{\text{geo}}(Q) = \{C_i, P_i, D_i, T_i(Q)\}_{i=1}^N. \quad (5)$$

C_i denotes the camera pose together with intrinsics derived from the predicted fields of view; P_i is the image-aligned point map expressed in the first-camera reference frame; D_i is depth along the rays of view i under the model's shared normalized scale; and $T_i(Q)$ contains tracks conditioned on queries rather than an unconditional scene tensor. Visibility and the confidence maps associated with depth and points are auxiliary outputs and are left outside the compact state notation. Normals, Gaussian parameters, semantics, or other variant-specific predictions are treated as derived outputs rather than part of the base geometric state.

The two components support different uses of the geometric state. $\mathcal{Z}_{\text{geo}}$ carries cross-view information before it is committed to a particular decoded output, whereas $\mathcal{R}_{\text{geo}}$ exposes cameras, points, depth, and tracks in an explicit coordinate frame. When states are produced from separate windows, their structured geometric outputs may additionally require pose, scale, and orientation alignment; whether the state is frozen, fine-tuned, adapted, distilled, or jointly trained describes how it is transferred rather than which component is used.

Fig. 3 summarizes the development of the VGGT [167] lineage around geometric state. The upper branches organize the 5 strengthening groups in Section 3, and the lower branches organize the 5 application groups in Section 4. Within each branch, methods are arranged by when they appeared.

2.4 Limitations of VGGT

The limitations of VGGT [167] fall into two groups. The first concerns the conditions under which its geometric state can be formed and retained for reconstruction, motivating the strengthening directions in Section 3. The second concerns how that state can support objectives beyond reconstruction, motivating the downstream reuse studied in Section 4.

On the reconstruction side, the lack of a general mechanism for handling optional geometry, additional sensors, and non-standard imaging motivates diverse-input 3D reconstruction; growing per-forward-pass cost motivates efficient 3D reconstruction; failures under degraded or ambiguous visual observations motivate robust 3D reconstruction; finite context and inconsistent predictions across windows motivate streaming and long-sequence 3D reconstruction; and the static-scene formulation motivates dynamic 3D reconstruction. Each direction strengthens how $\mathcal{Z}_{\text{geo}}$ is formed or retained, how it produces $\mathcal{R}_{\text{geo}}$, or how both components evolve over time.

On the downstream side, even accurate reconstruction leaves a reconstruction-oriented representation rather than a complete solution for later applications. The structured geometric outputs do not directly provide appearance, persistent mapping, semantic identity, future transitions, or action grounding, while the multi-view geometry latent features must still be selected and adapted for these objectives. Section 4 therefore examines how application models support NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Across these applications, the central issue shifts from improving state formation to selecting, adapting, or combining $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ according to the target capability.

3 Strengthening Geometric State for 3D Reconstruction

Reconstruction methods strengthen one or both components of VGGT's [167] geometric state. They improve how $\mathcal{Z}_{\text{geo}}$ is formed and retained, how reliably it produces $\mathcal{R}_{\text{geo}}$, or how both components extend over time. We organize these advances into diverse-input, efficient, robust, streaming and long-sequence, and dynamic 3D reconstruction. Table 1 maps these categories to the affected state components, mechanisms, and outcomes.

3.1 Diverse-Input 3D Reconstruction

VGGT [167] is designed primarily for pinhole RGB images and does not directly accommodate known geometry, additional sensors, or non-standard imaging systems. We therefore organize diverse-input reconstruction into 3 directions: optional camera and depth as input incorporates cameras, depth, or partial geometry; geometry-aware sensor fusion combines visual features with stereo or LiDAR measurements; and diverse imaging systems adapt reconstruction to high-resolution, fisheye, panoramic, or event inputs.

Optional camera and depth as input. Methods in this group differ in how optional camera, depth, or partial-geometry inputs are represented before multi-view fusion. OmniVGGT [127] injects encoded camera parameters and depth tokens, with stochastic training supporting different input combinations. MapAnything [78] factorizes calibration, poses, depth, and partial reconstructions into ray directions, ray depths, camera poses, and metric scale for a common VGGT-style [167] transformer. WorldMirror [104] jointly tokenizes intrinsics, poses, depth, and image features, then predicts several geometric and rendering outputs through task-specific heads. The key design choice is therefore whether optional geometry enters as additional tokens, factored geometric quantities, or a unified multi-task input representation.

Geometry-aware sensor fusion. StereoVGGT [24] and LiDAR-VGGT [171] extend the input setting beyond standard monocular RGB collections. Frozen VGGT [167] features provide useful camera-calibration knowledge for stereo, but direct use can lose the fine spatial detail needed for binocular matching. StereoVGGT [24] therefore applies a training-free feature-adjustment pipeline to restore spatial detail while exploiting the calibration information encoded in the frozen features. LiDAR-VGGT [171] couples LiDAR-inertial odometry with VGGT [167] through coarse-to-fine fusion, using metric sensing to constrain scale and large-environment consistency while retaining dense visual geometry.

Diverse imaging systems. Non-standard images require changes at different stages of reconstruction. HD-VGGT [21] separates coarse global inference from reliability-aware refinement to preserve high-resolution detail. For fisheye cameras, Fisheye3R [42] adapts reconstruction directly, whereas RayTun3R [147] corrects positional encodings biased toward pinhole cameras. Panoramic methods either reconstruct perspective slices and reproject them, as in VGGT-360 [197], or learn spherical encoding and rotation-aware training directly, as in PanoVGGT [60]. EventVGGT [135] changes the feature extraction stage for event-based depth. The required modification therefore depends on whether the input differs in resolution, camera projection, or sensing process.

The 3 groups broaden reconstruction through optional geometric information, additional sensor measurements, and diverse image formation, respectively. All expand the evidence available to form $\mathcal{Z}_{\text{geo}}$ and predict $\mathcal{R}_{\text{geo}}$, but none guarantees that every supplied measurement is preserved exactly in the final structured geometric outputs.

3.2 Efficient 3D Reconstruction

Dense multi-view processing in VGGT [167] requires substantial computation and memory as the number or resolution of input views grows. We therefore organize efficient reconstruction into

Table 1. Strengthening geometric state in 3D reconstruction. Different columns show what part of the state is strengthened, representative methods, how the strengthening is performed, and the resulting capability.

What is strengthened	Representative works	How it is strengthened	Result
Sec. 3.1 Diverse-Input 3D Reconstruction			
Z_{geo}: input evidence from optional geometry and additional sensors $Z_{\text{geo}}, \mathcal{R}_{\text{geo}}$: camera models and image-formation settings	(i) Optional camera and depth as input: OmniVGGT [127] / MapAnything [78] / WorldMirror [104] (ii) Geometry-aware sensor fusion: StereoVGGT [24] / LiDAR-VGGT [171] (iii) Diverse imaging systems: HD-VGGT [21] / Fisheye3R [42] / VGGT-360 [197] / PanoVGGT [60] / EventVGGT [135] / RayTun3R [147]	(i) Encode known cameras, depth, poses, or partial geometry (ii) Fuse stereo or LiDAR measurements with visual features (iii) Adapt feature extraction and geometric decoding to resolution, projection, or sensing modality	- Reconstruct from more input types - Use metric or partial-geometry constraints - Support non-standard cameras and imaging systems
Sec. 3.2 Efficient 3D Reconstruction			
Z_{geo}: token and feature computation $Z_{\text{geo}} \rightarrow \mathcal{R}_{\text{geo}}$: geometric information retained after compression	(i) Quantization-based compression: QuantVGGT [46] / VersaQ-3D [210] / TAPTQ [121] / FGQ [209] / QVGGT [124] (ii) Efficient network architecture design: FastVGGT [143] / LiteVGGT [145] / HTTM [176] / FlashVGGT [178] / Block-Sparse [165] / Co-Me [23] / AVGGT [154] / Attention Collapse [90] / HeSS [82] / VGGT-Ω [168] / PaceVGGT [91] / Spark3R [157] / Good Token Hunting [215] / RegimeVGGT [193] / TurboVGGT [69] / SAF3R [35] / Lite3R [203]	(i) Quantize weights and activations with sensitivity-aware precision (ii) Merge, prune, or select tokens and use sparse or compressed attention and compact heads	- Reduce runtime and memory per forward pass - Retain camera, depth, and reconstruction accuracy after compression
Sec. 3.3 Robust 3D Reconstruction			
Z_{geo}: evidence formed from degraded or contaminated images $Z_{\text{geo}} \rightarrow \mathcal{R}_{\text{geo}}$: reliability of predicted geometry	(i) Adverse environments: DarkVGGT [84] / Water-VGGT [105] / Wat3R [134] (ii) Distractor and outlier rejection: VGTW [122] / RobustVGGT [61]	(i) Add thermal observations, physical constraints, or domain adaptation (ii) Suppress distractors and reject irrelevant or non-overlapping views	- Recover more reliable geometry in darkness or underwater scenes - Prevent distractors and outlier views from corrupting reconstruction
Sec. 3.4 Streaming and Long-Sequence 3D Reconstruction			
Z_{geo}: historical information retained between windows $\mathcal{R}_{\text{geo}}$: scale and coordinate consistency across windows	(i) Causal reconstruction: StreamVGGT [224] / FrameVGGT [186] / STAC [173] / InfiniteVGGT [198] / OVGGT [107] / Evict3R [111] / XStreamVGGT [150] / StreamCacheVGGT [101] / RetrieveVGGT [225] / Mamba-VGGT [39] (ii) Long-sequence reconstruction: LASER [40] / LongStream [26] / Anchor3R [158] / VGGT-Long [36] / SwiftVGGT [87] / Diversity-aware Partitioning [125] / VGG-T³ [43]	(i) Select, compress, retrieve, or propagate history during causal updates (ii) Divide long inputs into local windows and align their scale and coordinates using anchors, loop closure, or specialized training	- Process causal streams and image collections larger than a single inference window - Limit memory while maintaining globally aligned cameras and geometry
Sec. 3.5 Dynamic 3D Reconstruction			
Z_{geo}: features that separate persistent structure from scene change $\mathcal{R}_{\text{geo}}$: time-varying geometry and motion	(i) Dynamic geometry and 4D representation: PAGE-4D [220] / 4D-VGGT [166] / V-DPM [151] / SelfEvo [71] / DynamicVGGT [63] (ii) Motion-guided geometry correction: VGGT4D [68] / Robust 4D VGGT [200] / 4DVGGT-D [201]	(i) Fuse information across space and time and predict dynamic or future geometry (ii) Detect moving regions and reduce their effect on camera, depth, and point predictions	- Stabilize camera and geometry estimates in dynamic scenes - Represent scene change and, where modeled, predict dense 3D motion

two directions: quantization-based compression reduces the numerical cost of model weights and activations, while efficient network architecture design reduces redundant token processing and attention computation.

Quantization-based compression. The central question is where reduced precision can be applied without damaging geometric prediction. QuantVGGT [46] and VersaQ-3D [210] identify activation outliers, unstable calibration, and heterogeneous precision requirements as the main obstacles to generic quantization. TAPTQ [121] compensates modules with long-tailed activation error, FGQ [209] allocates precision using task-, block-, and channel-level Fisher sensitivity, and QVGGT [124] combines block-wise mixed precision with special-token filtering and consistency-aware scale search. The progression is from uniform compression toward sensitivity-aware precision allocation around the pathways responsible for camera, depth, and point predictions.

Efficient network architecture design. Beyond quantization-based compression, a second line reduces architectural redundancy. FastVGGT [143], LiteVGGT [145], Co-Me [23], and PaceVGGT [91] merge or prune tokens using similarity, geometric confidence, or learned pre-attention importance. Spark3R [157], Good Token Hunting [215], and RegimeVGGT [193] distinguish query from key-value roles, frame from patch selection, or shallow, middle, and deep layer regimes. HTTM [176], FlashVGGT [178], block-sparse global attention [165], HeSS [82], TurboVGGT [69], and SAF3R [35] instead restructure or adapt attention across heads, layers, and inputs; Lite3R [203] combines sparse linear attention with low-precision training. Analyses by AVGGT [154] and the attention-collapse study [90] explain why such reductions must remain selective, while VGGT- Ω [168] uses a unified dense head and scene tokens for feature aggregation. The shared logic is to reduce token processing and attention computation across heads, layers, and views while retaining the cross-view geometric information needed by the prediction heads.

The two directions reduce the cost of a forward pass in different ways: quantization lowers numerical cost, whereas efficient network architecture design reduces token processing and attention computation. Both aim to form $\mathcal{Z}_{\text{geo}}$ more efficiently without losing the cross-view information needed for stable $\mathcal{R}_{\text{geo}}$.

3.3 Robust 3D Reconstruction

VGGT [167] can become unreliable when visual evidence is degraded by the environment or contaminated by distractors and inconsistent views. We therefore organize robust reconstruction into two directions: methods for adverse environments recover or supplement weak visual evidence, while distractor and outlier rejection prevents irrelevant content or views from corrupting multi-view fusion.

Adverse environments. The appropriate robustness mechanism depends on why visual evidence is degraded. DarkVGGT [84] supplements missing visible-light structure with RGB-thermal observations. Underwater reconstruction instead addresses attenuation and scattering: WaterVGGT [105] imposes water-optics constraints, while Wat3R [134] adapts geometry from unlabeled underwater video through cross-view consistency. Thermal sensing provides additional observations in darkness, whereas physical constraints and adaptation address systematic environmental distortion.

Distractor and outlier rejection. VGTW [122] suppresses transient distractors and occlusions during cross-view fusion. RobustVGGT [61] addresses irrelevant or non-overlapping input views by exploiting emergent outlier suppression in VGGT [167] attention and intermediate features. It filters inconsistent views with a global threshold and reruns reconstruction without fine-tuning. Both methods improve robustness to contaminated image sets by identifying inputs that should contribute less to $\mathcal{Z}_{\text{geo}}$.

The two groups address different causes of unreliable input evidence: adverse environments degrade otherwise relevant observations, whereas input contamination introduces views or content that should be rejected. Existing methods improve the reliability of $\mathcal{R}_{\text{geo}}$ by supplementing weak evidence or suppressing inconsistent evidence. Coverage remains condition-specific: darkness,

underwater scenes, and distractors have dedicated solutions, whereas rain, fog, and compound weather corruption remain less studied within the current VGGT [167] literature.

3.4 Streaming and Long-Sequence 3D Reconstruction

VGGT [167] jointly processes a bounded image set and therefore cannot update its reconstruction causally as new frames arrive or maintain a globally consistent reconstruction over long sequences. We therefore organize this subsection into two directions: causal reconstruction updates geometry online as each frame arrives, while long-sequence reconstruction connects bounded local predictions into a consistent global result.

Causal reconstruction. The first direction addresses online reconstruction with an unknown sequence length: the model must update the current geometry before seeing future observations. StreamVGGT [224] introduces causal attention with a historical key-value (KV) cache, FrameVGGT [186] organizes bounded context into frame-wise units, and STAC [173] compresses past information using temporal salience and voxel-aligned spatial aggregation. InfiniteVGGT [198], OVGGT [107], and Evict3R [111] bound the retained context through rolling selection, anchor protection, or training-free eviction. XStreamVGGT [150] further combines KV pruning and quantization, while StreamCacheVGGT [101] selectively merges cached tokens to reduce cache size. RetrieveVGGT [225] rebuilds a fixed context through retrieval, and Mamba-VGGT [39] propagates a compact external state. These designs strengthen $\mathcal{Z}_{\text{geo}}$ by selecting, compressing, retrieving, or propagating the past evidence most useful for each causal update.

Long-sequence reconstruction. The second direction addresses long inputs whose observations cannot be processed in one bounded forward pass. Individually accurate local predictions can become mutually inconsistent after many state updates, so long-sequence methods must both partition observations and reconnect their outputs. LASER [40] aligns layer-dependent scales across adjacent windows, while LongStream [26] learns keyframe-relative poses, orthogonal scale, cache consistency, and periodic refresh. Anchor3R [158] avoids a persistent first-frame gauge: it predicts current-centric local measurements and uses transient anchors, motion averaging, and loop-closure reinsertion to recover a global map. VGGT-Long [36] combines overlapping chunks with lightweight loop closure, while SwiftVGGT [87] detects loops from backbone features and applies single-step Sim(3) alignment. Diversity-aware View Partitioning [125] balances visual dissimilarity and spatial dispersion when constructing chunks. VGG-T³ [43] distills variable-length KV geometry into a fixed-size multilayer perceptron (MLP) state through test-time training. These methods strengthen $\mathcal{R}_{\text{geo}}$ by keeping decoded cameras and geometry in a consistent scale and coordinate system as the sequence grows.

Together, the two directions address online state updates and globally consistent reconstruction over inputs that exceed a bounded inference window. Causal reconstruction retains the evidence needed for each update, while long-sequence reconstruction aligns the resulting local predictions across the full sequence.

3.5 Dynamic 3D Reconstruction

VGGT [167] assumes a static scene, so moving content can corrupt camera and geometry estimates and cannot be represented as time-varying structure. We therefore organize dynamic 3D reconstruction into two directions: the dynamic geometry and 4D representation direction explicitly models scene structure and motion over time, while motion-guided geometry correction identifies dynamic or uncertain regions to stabilize pose and geometry.

Dynamic geometry and 4D representation. The first direction represents scene changes within the geometric state rather than only correcting their effect on static reconstruction. PAGE-4D [220] uses dynamics-aware masks to route motion evidence differently for pose and geometry,

and 4D-VGGT [166] factorizes cross-view spatial fusion and cross-time local fusion. V-DPM [151] makes the extension explicit with time-varying dynamic point maps and dense 3D motion, together with camera and geometry prediction in the full model. SelfEvo [71] addresses limited 4D supervision: its spatio-temporal self-distillation adapts pretrained reconstruction models, including VGGT [167], using unlabeled videos rather than scarce 4D annotations. Driving scenes provide a demanding instance of this technical direction because they contain many moving objects and require temporally consistent geometry over long trajectories. DynamicVGGT [63] addresses this setting by jointly predicting current and future point maps, applying motion-aware temporal attention, and modeling point motion with a dynamic Gaussian head.

Motion-guided geometry correction. The second direction retains the pretrained VGGT representation but reduces the effect of moving regions on VGGT [167] predictions. VGGT4D [68] mines layer-wise motion cues from global attention and feeds refined dynamic masks back into early inference. Robust 4D VGGT [200] combines attention-derived uncertainty with local geometry purification, whereas 4DVGGT-D [201] first stabilizes pose against dynamic interference and then refines static and dynamic depth. These training-free methods detect motion or uncertainty and use it to correct pose, depth, or point-map predictions. The two directions respectively represent scene change within the geometric state and correct base geometry using motion cues.

Dynamic variants make both forms of geometric information time dependent: the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$ must distinguish persistent structure from change, while the structured geometric outputs $\mathcal{R}_{\text{geo}}$ expose time-varying geometry and, where modeled, motion without destabilizing the coordinate frame.

3.6 Summary of Geometric State Strengthening

This section has shown how geometric state is strengthened through broader inputs, lower computation, greater robustness, retained streaming context, and explicit scene dynamics. These directions are not independent: streaming and long-sequence reconstruction depend on efficient computation, while online reconstruction of dynamic scenes also requires memory mechanisms that preserve context without treating motion as static structure. Combining multiple strengthening directions therefore broadens both the conditions under which $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ remain useful and the information that later applications can obtain from the state. Section 4 next examines those applications.

4 Reusing Geometric State Beyond 3D Reconstruction

Downstream applications reuse the structured geometric outputs $\mathcal{R}_{\text{geo}}$, the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$, or their combination. We organize this literature into 5 applications: NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Table 2 summarizes the geometric-state information used in each application and the corresponding method groups.

4.1 Novel View Synthesis

Novel-view synthesis converts the geometric state into a renderable scene representation for synthesizing unseen views. With 3D Gaussian Splatting [80] as the dominant representation, methods differ in whether feed-forward geometry completes the reconstruction or provides a starting point for later refinement. We therefore organize them into direct feed-forward 3D Gaussian reconstruction and 3D Gaussian reconstruction initialized by feed-forward geometry.

Direct feed-forward 3D Gaussian reconstruction. This route decodes the Gaussian scene without first optimizing a separate representation from structured geometric outputs. AnySplat [75], VGD [96], and SplatWeaver [163] train feed-forward Gaussian heads with VGGT [167]-derived

Table 2. Geometric-state reuse across 5 downstream applications: novel view synthesis (NVS), SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning. Different columns show which geometric-state components and information are used in each application and list the corresponding method groups.

Geometric-state components and information used	Method groups and representative works
Sec. 4.1 Novel View Synthesis	
$\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used to construct or initialize renderable geometry $\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features decoded into appearance or Gaussian parameters	Direct feed-forward 3D Gaussian reconstruction: AnySplat [75] / VGD [96] / SplatWeaver [163] / G3Splat [66] / ReconDrive [195] / OF³GS [19] 3D Gaussian reconstruction initialized by feed-forward geometry: VGGT-X [103] / Gesplat [106] / VGGs [181] / Geo-EVS [85] / Panolmager [187] / SSR-GS [44] / Distractor-Free GS [56] / AsyncEvGS [31] / RGB-Thermal GS [28]
Sec. 4.2 SLAM	
$\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used for tracking, alignment, and mapping $\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features used for correspondence, retrieval, and place recognition	SLAM systems: VGGT-SLAM [110] / VGGT-SLAM 2.0 [109] / VGGT-SLAM++ [112] / EC3R-SLAM [67] / AIM-SLAM [74] / VGGT-Geo [130] / VGGT-Mapping [156] / CALM [206] Visual odometry: VGGT-Motion [183] / HyVGGT-VO [120] / Keyframe VO [32] Relocalization: VGGT-MPR [184] / UniPR-3D [37] / GPA-VGGT [185] / Reloc-VGGT [38] / MultiLoc [33] / RIC-Loc [77] / G2IA [76] / UAV Geometric Localization [92] / GeoLink [202]
Sec. 4.3 Semantic 3D Scene Understanding	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features used for segmentation, language reasoning, and association $\mathcal{R}_{\text{geo}}$: 3D locations and geometry used to ground semantic predictions	3D semantic prediction and mapping: SegVGGT [131] / VGGT-Segmentor [49] / VGGT-Det [15] / OV3DSeg-VGGT [218] / VGGT-Occ [22] / GPOcc [217] / DeGO [48] / Dense Semantic Matching [189] / MuViSeg [45] / SceneVGGT [53] / VGGT-CD [207] Language-based 3D scene reasoning: MVGGT [179] / 4DLangVGGT [180] / SpaceMind [212] / 3DThinker [25] / Geometric Distillation [88] / Distilled 3D VLM [4] / DR-MV3D [27] / EAGLE [14]
Sec. 4.4 Geometry-Aware World Models	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features predicted over time or used to condition and supervise generation $\mathcal{R}_{\text{geo}}$: cameras, depth, and point maps used to condition or evaluate future predictions	Geometry-conditioned video-depth prediction: VGGT-World [153] Geometry-conditioned video generation: CETCAM [214] / CineScene [70] / CamGeo [100] / WristWorld [129] / VideoGPA [41] / V-Dreamer [62] Geometry-guided action prediction and rollout selection: MECo-WAM [205] / GEM-4D [219] / Test-Time Geometric Scaling [213]
Sec. 4.5 Embodied Action and Planning	
$\mathcal{Z}_{\text{geo}}$: multi-view geometry latent features fused into policies or geometry-aware VLA models $\mathcal{R}_{\text{geo}}$: depth, points, or camera geometry used for planning and feasible action	Geometry-enhanced action policies: VGGT-DP [50] / R3DP [211] / VO-DP [117] / Efficient Geometry Encoders [162] / OCRA [170] Geometry-based planning and trajectory generation: GraspView [174] / SyncTwin [72] / MG-Nav [164] Geometry-aware VLA models: 3D-Mix [194] / GeoAware-VLA [1] / GLaD [59] / AugVLA-3D [132] / ABot-M0 [191] / VGGDrive [169] / Controlled Analysis [190]

initialization or distillation; SplatWeaver further distills camera and depth supervision while allocating primitives adaptively. G3Splat [66] regularizes the geometry of the decoded Gaussians, ReconDrive [195] adds spatio-temporal heads for dynamic driving scenes, and OF³GS [19] incrementally predicts, refines, and fuses Gaussians from a frozen StreamVGGT [224] front end. These variants trade a separate scene-optimization stage for direct or incremental inference.

3D Gaussian reconstruction initialized by feed-forward geometry. This route retains a subsequent reconstruction or optimization stage after cameras, points, or depth have been inferred. VGGT-X [103] scales and globally aligns VGGT [167] outputs before robust 3DGS [80] training; Gesplat [106] and VGGs [181] initialize pose-free or sparse-view reconstruction from cameras, points, or depth. The same principle extends to extrapolative driving views in Geo-EVS [85], sparse panoramic views in Panolmager [187], glossy surfaces in SSR-GS [44], and transient-content removal in Distractor-Free GS [56]. AsyncEvGS [31] and unpaired RGB-thermal GS [28] additionally use the inferred geometry to align event or thermal observations. Feed-forward structured geometric

outputs provide the geometric starting point, while the retained reconstruction stage refines appearance and adapts to the target setting.

The distinction is whether Gaussians are reconstructed directly from the feed-forward representation or refined after initialization from $\mathcal{R}_{\text{geo}}$. Both routes must preserve cross-view structure when appearance, dynamics, or sensing modality changes.

4.2 SLAM

SLAM uses VGGT [167] or its long-sequence extensions as geometric front ends to build consistent maps and camera trajectories. Unlike the reconstruction methods reviewed in Section 3.4, SLAM adds the system components needed for joint mapping and tracking, camera-motion estimation without full mapping, and pose recovery in previously observed scenes. We therefore organize these methods into SLAM systems, visual odometry, and relocalization.

SLAM systems. The first group uses VGGT [167] as a dense front-end and adds the tracking, submap alignment, and global optimization needed to construct a consistent map. VGGT-SLAM [110], VGGT-SLAM 2.0 [109], and VGGT-SLAM++ [112] align feed-forward submaps with projective or global optimization. EC3R-SLAM [67] and AIM-SLAM [74] reduce redundant inference through sparse tracking or informative keyframes, while VGGT-Geo [130] adds confidence-aware geometric constraints and VGGT-Mapping [156] supports robot navigation. CALM [206] extends this role to kilometer-scale mapping by using an assistant camera baseline, intrinsic correction, and anchor propagation to resolve scale and nonlinear submap distortion. Across these systems, $\mathcal{R}_{\text{geo}}$ is useful only after local gauges and accumulated drift are reconciled.

Visual odometry. Unlike SLAM systems, these methods focus on estimating camera motion without constructing and globally optimizing a persistent map. VGGT-Motion [183] and HyVGGT-VO [120] couple dense geometry to long-range camera tracking or visual odometry, while keyframe-based feed-forward VO [32] learns which frames to include in the VGGT [167] window. Per-window pose accuracy alone is insufficient: temporal selection, scale alignment, and global consistency determine whether small local errors accumulate along the estimated trajectory.

Relocalization. Place-recognition methods first identify relevant references: VGGT-MPR [184] uses VGGT [167] embeddings, depth-assisted LiDAR densification, and tracking-based re-ranking; UniPR-3D [37] aggregates 2D and 3D VGGT [167] tokens for multi-view retrieval; G2IA [76] combines VGGT [167] geometry priors with instance-aware candidate verification; and GeoLink [202] distills offline VGGT [167] geometry into a 2D-only cross-view retrieval representation. Pose relocalization is addressed by GPA-VGGT [185], Reloc-VGGT [38], and MultiLoc [33] through VGGT [167] adaptation or multi-reference pose estimation, while RIC-Loc [77] derives multiple posed-reference hypotheses and uses their consensus and dispersion for selective localization. UAV geometric localization [92] reconstructs local 3D structure and renders a bird’s-eye-view intermediary for cross-view 3-DoF pose regression. This route is hybrid: $\mathcal{Z}_{\text{geo}}$ narrows or transfers the search representation, while $\mathcal{R}_{\text{geo}}$ establishes geometric compatibility.

SLAM therefore adds 3 capabilities to feed-forward or streaming geometry: globally consistent mapping, visual odometry, and recovery of pose in a previously observed scene. Structured geometric outputs anchor geometry and pose, while multi-view geometry latent features support correspondence and retrieval.

4.3 Semantic 3D Scene Understanding

Semantic 3D scene understanding changes the target from reconstructing structure to attaching categories, language, and persistent identities to it. Semantic heads read multi-view spatial features from $\mathcal{Z}_{\text{geo}}$, while $\mathcal{R}_{\text{geo}}$ anchors their predictions in 3D. We therefore organize these methods into two directions: 3D semantic prediction and mapping; and language-based 3D scene reasoning.

3D semantic prediction and mapping. This group predicts semantic labels for objects, visible regions, or occupied 3D space and maintains those predictions across views or time. SegVGGT [131], VGGT-Segmentor [49], and VGGT-Det [15] address instance segmentation, cross-view segmentation, or multi-view detection. OV3DSeg-VGGT [218] extends segmentation to open-vocabulary categories. Beyond visible surfaces, VGGT-Occ [22] decodes semantic occupancy from a frozen multi-view encoder, GPOcc [217] converts surface points and features into sparse Gaussian occupancy, and DeGO [48] distills cross-view and cross-time features into deformable Gaussians. Persistent semantic mapping extends these outputs across views or time: Dense Semantic Matching [189] associates pixels across object instances, MuViSeg [45] matches whole segments jointly across multiple views, SceneVGGT [53] retains instance identities and timestamps in an online 3D map, and VGGT-CD [207] registers observations from different epochs to identify physical scene changes.

Language-based 3D scene reasoning. Methods differ in whether geometry enters through explicit grounding, feature fusion, or distillation. MVGGT [179] and 4DLangVGGT [180] ground referring expressions across views or time. SpaceMind [212] fuses VGGT [167] features under camera guidance, whereas 3DThinker [25] aligns a reasoning latent with VGGT [167] geometry. Geometric Distillation [88] transfers correspondence, relative-depth, and cost-volume cues into a vision-language model (VLM), and Distilled 3D VLM [4] adopts VGGT [167] as the encoder of a compact reasoning model. DR-MV3D [27] supervises global-map construction and view planning with geometry-consistent pseudo targets. EAGLE [14] localizes visual queries in 3D by combining episodic appearance and geometry memories. Each mechanism connects language or query outputs to cross-view spatial relations, but at a different stage of the reasoning model.

Together, the two directions progress from assigning meaning to 3D structure to reasoning about that structure with language or query memories. They typically combine transferable multi-view geometry latent features in $\mathcal{Z}_{\text{geo}}$ with the spatial anchors provided by structured geometric outputs in $\mathcal{R}_{\text{geo}}$.

4.4 Geometry-Aware World Models

Geometry-aware world models differ in what they predict and where geometric state enters the prediction process. Geometry can be predicted as future depth, guide the generation of future visual observations, or support action prediction and the selection of geometrically consistent rollouts. We therefore organize these methods into geometry-conditioned video-depth prediction, geometry-conditioned video generation, and geometry-guided action prediction and rollout selection.

Geometry-conditioned video-depth prediction. VGGT-World [153] takes RGB observations as input and predicts future video depth by evolving $\mathcal{Z}_{\text{geo}}$ autoregressively rather than predicting future pixels alone. The model uses the current multi-view geometry latent features to forecast a future geometric state, and structured geometric outputs then decode or verify the depth and camera geometry implied by the predicted state. This setting directly tests whether geometric state can support temporally coherent depth prediction beyond the observed frames.

Geometry-conditioned video generation. Geometry can condition generation as explicit tokens, multi-view geometry latent features, distilled supervision, or an evaluation signal. CET-CAM [214] tokenizes predicted cameras and depth, whereas CineScene [70] injects multi-view geometry latent features. CamGeo [100] distills camera, depth, and cross-frame constraints during training. WristWorld [129] reconstructs poses and 4D point maps before generating future wrist-camera views, while VideoGPA [41] forms preference pairs from a VGGT [167] reconstruction score. V-Dreamer [62] uses video generation priors to synthesize robot interaction videos and then converts their motion into kinematically aligned trajectories. Although the resulting trajectories support planning, its core predictive mechanism is video generation, so we classify it as a world

model. The relevant evidence is whether each form of generation preserves coherent structure under camera, scene, or robot motion and yields useful downstream trajectories where applicable.

Geometry-guided action prediction and rollout selection. MECo-WAM [205] transfers relational 4D targets from a frozen VGGT [167] encoder into a video-action world model. GEM-4D [219] distills dense correspondence from a pretrained 4D geometry model into video prediction and converts consistent rollouts into robot trajectories; VGGT [167] is evaluated as a teacher choice rather than assumed to be its default geometry model. Test-Time Geometric Scaling [213] uses frozen VGGT- Ω [168] camera, depth, and point-map predictions to rank sampled world-action rollouts by reprojection consistency and selectively spend additional computation. These methods use geometry to supervise action prediction or select a geometrically consistent rollout; their value must ultimately appear in both geometric consistency across predicted frames and improved action success.

Visual fidelity alone is insufficient for these methods. Future observations must preserve 3D structure, and action-conditioned models must show that this structure improves an action prediction or reliably identifies a better rollout.

4.5 Embodied Action and Planning

Embodied methods use geometric state to choose, plan, or execute actions that affect an environment. Different action models incorporate geometry at different stages, from policy features to explicit motion planning and multimodal action reasoning. We therefore organize them into geometry-enhanced action policies, geometry-based planning and trajectory generation, and geometry-aware VLA models.

Geometry-enhanced action policies. The design choice is how geometric evidence enters action prediction. VGGT-DP [50] and R3DP [211] fuse multi-view geometry latent features with the policy, VO-DP [117] balances semantic and geometric observations, and Efficient Geometry Encoders [162] reduce the latency of this transfer. OCRA [170] adds object-centric point clouds and tactile priors instead of relying on dense features alone. In these policies, $\mathcal{Z}_{\text{geo}}$ provides multi-view spatial features and selected structured geometric outputs provide explicit geometry for contact-related action prediction.

Geometry-based planning and trajectory generation. These methods use reconstructed geometry directly to select views, plan motion, or generate feasible trajectories. GraspView [174] uses reconstruction for view selection and grasp alignment, SyncTwin [72] maintains a registered digital twin, and MG-Nav [164] aligns observations and goals in spatial memory. Structured geometric outputs, alone or together with multi-view geometry latent features, connect observations and memory to feasible motion in these systems.

Geometry-aware VLA models. These methods add geometric information to vision-language-action (VLA) models that connect visual observations and language to actions. 3D-Mix [194] and GeoAware-VLA [1] use plug-and-play or implicit feature fusion, while GLaD [59] distills VGGT [167] features into the visual tokens of a VLA model. AugVLA-3D [132] augments action features with depth cues derived from VGGT [167]. ABot-M0 [191] admits VGGT [167] as one optional plug-and-play geometry module in a dual-stream policy, rather than making it the policy's core. VGGDrive [169] injects cross-view features into a driving VLM for risk perception, motion prediction, and trajectory planning, rather than direct low-level control. Controlled analysis [190] compares several VGGT [167]-VLA fusion strategies under matched settings. These models mainly use multi-view geometry latent features and retain structured geometric outputs when contact, collision, or feasibility requires geometric constraints.

Together, these directions use geometric state at 3 levels of embodied decision making: learned action prediction, explicit planning, and multimodal action reasoning. Multi-view geometry latent

Table 3. Evaluation-task coverage within the dataset collection summarized in Table 4. Each cell reports the number and percentage of datasets that provide observations or annotations supporting the corresponding evaluation target, with background color encoding the percentage.

Dataset group	Dataset count	State-strengthening tasks					Downstream-reuse tasks				
		Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. under.	World models	Emb. action/plan.
3D reconstruction	46	36 (78%)	35 (76%)	24 (52%)	18 (39%)	15 (33%)	8 (17%)	20 (43%)	15 (33%)	5 (11%)	0 (0%)
NVS	10	4 (40%)	1 (10%)	1 (10%)	0 (0%)	2 (20%)	10 (100%)	0 (0%)	0 (0%)	2 (20%)	0 (0%)
Other downstream	15	1 (7%)	2 (13%)	1 (7%)	0 (0%)	2 (13%)	0 (0%)	0 (0%)	5 (33%)	8 (53%)	10 (67%)

Coverage 0% 100%

features support transferable policy representations, while structured geometric outputs provide the spatial precision needed to evaluate feasible motion and interaction.

4.6 Summary of Downstream State Reuse

Across the 5 downstream applications, the required access to geometric state depends on the capability being built. NVS and SLAM rely strongly on structured geometric outputs, semantic applications attach meaning to spatial anchors, world models require representations that support transitions beyond observed evidence, and embodied methods combine geometric state with action objectives. Although this is not a strict progression, applications closer to prediction and closed-loop action tend to combine the geometric precision of $\mathcal{R}_{\text{geo}}$ with the transferable information in $\mathcal{Z}_{\text{geo}}$. The common requirement is therefore not one fixed way of using the state, but preservation of the information needed by each application under its supervision and temporal setting; Section 5 next examines the evidence used to evaluate these capabilities.

5 Datasets and Benchmarks

This section reviews the datasets and evaluation targets used for geometric-state strengthening and downstream reuse. Section 5.1 groups datasets by their primary use in 3D reconstruction, NVS, or other downstream tasks. Sections 5.2 and 5.3 then examine how state-strengthening methods and downstream applications are evaluated, and Section 5.4 summarizes the limitations of current datasets and evaluation targets.

5.1 Dataset Taxonomy

We organize evaluation resources into 3 groups according to their primary use: datasets for 3D reconstruction, datasets for novel-view synthesis (NVS), and datasets for other downstream tasks. The assignment is not exclusive; a dataset remains in its principal group even when its observations or annotations support the evaluation of additional tasks. The collection includes datasets used by the reviewed methods and established resources for the corresponding task families. Table 3 first summarizes task-support coverage within this collection. Table 4 then provides dataset-level details, including scene type, source, reported dataset scale, and supported strengthening or reuse tasks.

Datasets for 3D reconstruction. The evolution of reconstruction datasets can be traced through 3 representative stages. DTU [73] established controlled multi-view evaluation on object-scale scans; ETH3D [141] later extended accurate geometry to high-resolution indoor and outdoor

scenes; and Waymo [152] now represents large-scale, multimodal outdoor data with broad task annotations. This evolution proceeds along 3 directions. First, spatial coverage expands from isolated objects to rooms, landmarks, and city-scale environments, as illustrated by ScanNet v2 [30], ScanNet200 [137], MegaDepth [94], and Waymo [152]. Second, temporal coverage moves from bounded image sets to trajectories and long sequences, with KITTI [52], TartanAir [177], and Waymo [152] supporting trajectory estimation and long-sequence reconstruction evaluation. Third, sensing conditions become more diverse: Replica [148] and Kubric [55] provide exact synthetic geometry and controllable motion, while real collections increasingly include changing illumination, non-pinhole cameras, events, thermal sensing, and LiDAR. Within the dataset collection summarized in Table 3, this group provides broader support for camera pose, depth, reconstruction, long-stream, dynamic, and SLAM evaluation.

Datasets for NVS. NVS benchmarks likewise progress from established video collections to larger and more diverse scene collections. RealEstate10K [221] provided an early large collection of posed real-estate videos; Mip-NeRF 360 [6] later established evaluation for unbounded scenes; and DL3DV-10K [97] now supplies large-scale real-world videos spanning indoor and outdoor environments. Their development follows 3 directions. First, scene coverage broadens across object-centric, outdoor, and unbounded settings: CO3Dv2 [133] covers 51 object categories through multi-view object-centric sequences, ACID [98] focuses on outdoor scenes, and Mip-NeRF 360 [6] evaluates unbounded environments. Second, temporal complexity extends static-scene rendering to dynamic content through DyCheck [47] and HyperNeRF [126]. Third, data scale increases from a small number of carefully captured scenes, such as Deep Blending [64] and LLFF [114], to large real-video collections such as RealEstate10K [221] and DL3DV-10K [97]. Table 3 shows that all datasets in this group support NVS evaluation, while fewer support depth, reconstructed geometry, temporal consistency, or downstream tasks.

Datasets for other downstream tasks. This group is organized by the applications being evaluated. Semantic datasets range from urban scene understanding in Cityscapes [29] to referring expressions in MVRefer [179], interaction-aware 4D understanding in HOI4D [102], and aligned egocentric and exocentric observations in Ego-Exo4D [54]. World-model datasets support interaction-conditioned or controllable future generation through HoI HQ [214] and Scene-Decoupled-Video-dataset [70]. Embodied datasets progress from controlled manipulation suites such as MetaWorld [196], CALVIN [113], ManiSkill 3 [159], LIBERO [99], and RoboTwin 2.0 [20] to real-world interaction and driving data in DROID [81], Ego-Exo4D [54], and NAVSIM [34]. Cityscapes [29] provides 5,000 urban video snippets, while NAVSIM [34] expands task-oriented evaluation to 115,000 filtered driving scenarios. Table 3 consequently shows moderate coverage for semantic understanding, broader coverage for world models, and the strongest coverage for embodied action, but little support for NVS or SLAM evaluation targets; these datasets measure application outcomes well but provide less evidence about whether gains arise specifically from improved geometric state.

5.2 Evaluation Targets for Geometric State Strengthening

Evaluation targets for diverse-input, efficient, robust, streaming and long-sequence, and dynamic reconstruction often assess the same geometric outputs under different conditions. Diverse-input methods vary the available measurements, efficient methods vary per-forward-pass precision and computation, and robust methods vary sensing conditions or input corruption. Streaming and long-sequence methods evaluate state transfer and alignment across windows, while dynamic methods evaluate time-varying scenes. Geometric accuracy must therefore be reported together with the condition being tested. Table 5 summarizes common geometric measures and the evaluation focus, representative datasets, key metrics, and comparison conditions for each strengthening direction.

Table 4. Datasets included in the evaluation taxonomy, grouped by primary use and supported tasks. A ✓ indicates that the dataset provides observations or annotations supporting the corresponding evaluation target; an ✗ indicates that such support was not identified. #Scenes reports the dataset scale used in this table; alternative reported scales are shown in parentheses when a scene count is unavailable. Type symbols denote  objects,  indoor scenes,  outdoor scenes, and  indoor and outdoor scenes. Source identifies real, synthetic, or mixed data.

Dataset	#Scenes	Type	Source	State-strengthening tasks					Downstream-reuse tasks				
				Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. Under.	World models	Emb. action/plan.
Datasets for 3D Reconstruction													
DTU [73]	124		Real	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗
TUM-dynamics [149]	1		Real	✓	✓	✗	✓	✓	✗	✓	✗	✗	✗
ADT [123]	2		Real	✓	✓	✓	✗	✓	✗	✗	✓	✗	✗
EuRoC MAV [9]	3		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
TUM RGB-D [149]	4		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
7-Scenes [144]	7		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
NRGBD [5]	10		Synthetic	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗
Replica360-2K [136]	13		Synthetic	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
Replica [148]	18		Synthetic	✓	✓	✓	✗	✗	✓	✓	✓	✗	✗
Bonn [119]	24		Real	✓	✓	✓	✓	✓	✗	✓	✗	✗	✗
Matterport3D [17]	90		Real	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
Dark3R [58]	104		Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
Stanford2D3D [3]	271		Real	✗	✓	✓	✗	✗	✗	✗	✓	✗	✗
NYU-v2 [146]	464		Real	✗	✓	✗	✗	✗	✗	✗	✓	✗	✗
ScanNet++ v2 [192]	1,006		Real	✓	✓	✓	✗	✗	✓	✓	✓	✗	✗
ScanNet v2 [30]	1,513		Real	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗
ScanNet200 [137]	1,513		Real	✓	✓	✓	✗	✗	✗	✓	✓	✗	✗
ARKitScenes [7]	1,661		Real	✓	✓	✓	✗	✗	✗	✗	✓	✗	✗
EventScape [51]	4		Synthetic	✗	✓	✗	✗	✓	✗	✗	✓	✗	✗
Nordland [155]	4		Real	✗	✗	✗	✓	✗	✗	✓	✗	✗	✗
Cambridge Landmarks [79]	5		Real	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗
Virtual KITTI 2 [11]	5		Synthetic	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗
Oxford Spires [160]	6		Real	✓	✗	✓	✓	✗	✓	✓	✗	✗	✗
DENSE [65]	8		Synthetic	✗	✓	✗	✗	✗	✗	✗	✗	✗	✗
KITTI-360 [95]	11		Real	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
PointOdyssey [216]	20		Synthetic	✓	✓	✗	✓	✓	✗	✗	✗	✗	✗
KITTI Odometry [52]	22		Real	✓	✓	✓	✓	✗	✗	✓	✗	✗	✗
TartanAir [177]	30		Synthetic	✓	✓	✓	✓	✓	✗	✓	✓	✓	✗
MegaDepth [94]	196		Real	✗	✓	✓	✗	✗	✗	✗	✗	✗	✗
nuScenes [13]	1,000		Real	✓	✓	✗	✗	✓	✓	✗	✓	✓	✗
Waymo [152]	1,150		Real	✓	✓	✗	✓	✓	✓	✗	✓	✓	✗
Oxford RobotCar [108]	– (~20M frames) ^a		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
LaMAR [138]	3		Real	✓	✗	✗	✗	✗	✗	✓	✗	✗	✗
STheReO [199]	3		Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
FIORD [57]	10		Real	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
MVSEC [223]	11		Real	✓	✓	✗	✓	✗	✗	✓	✗	✗	✗
BEDLAM2.0 [161]	15		Synthetic	✓	✓	✗	✗	✓	✗	✗	✗	✗	✗
Tanks and Temples [83]	21		Real	✗	✗	✓	✗	✗	✓	✗	✗	✗	✗
ETH3D [141]	25		Real	✓	✓	✓	✗	✗	✗	✗	✗	✗	✗
NCLT [16]	27		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
TUM-VI [142]	28		Real	✓	✗	✗	✓	✗	✗	✓	✗	✗	✗
Sintel [10]	35		Synthetic	✓	✓	✗	✗	✓	✗	✗	✗	✗	✗

^aFor Oxford RobotCar [108], OmniWorld [222], and SPair-71k [116], we did not identify a reported scene count in the source used here; frame or pair counts are shown to indicate dataset scale.

Within bounded reconstruction, diverse-input targets vary intrinsics, poses, depth, partial geometry, sensing modality, or camera model and test whether these inputs improve camera, depth,

TABLE 4 (CONTINUED)

Dataset	#Scenes	Type	Source	State-strengthening tasks					Downstream-reuse tasks				
				Cam. pose	Depth	3D recon.	Long stream	Dyn. 4D	NVS	SLAM	Sem. Under.	World models	Emb. action/plan.
Datasets for 3D Reconstruction (continued)													
ViViD++ [86]	36	 	Real	✓	✓	✗	✗	✗	✗	✗	✗	✗	✗
DAVIS [128]	50	 	Real	✗	✗	✗	✗	✓	✓	✗	✓	✗	✗
Kubric [55]	10,000	 	Synthetic	✓	✓	✓	✗	✓	✗	✗	✗	✗	✗
OmniWorld [222]	— (18.515M frames)	 	Real + Synthetic	✓	✓	✓	✗	✓	✗	✗	✗	✓	✗
Datasets for NVS													
CO3Dv2 [133]	51		Real	✓	✗	✓	✗	✗	✓	✗	✗	✗	✗
NVS-HO [2]	67		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Deep Blending [64]	19		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
ACID [98]	891		Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
DyCheck [47]	7	 	Real	✓	✓	✗	✗	✓	✓	✗	✗	✗	✗
LLFF [114]	8	 	Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
Mip-NeRF 360 [6]	9	 	Real	✗	✗	✗	✗	✗	✓	✗	✗	✗	✗
HyperNeRF [126]	10	 	Real	✗	✗	✗	✗	✓	✓	✗	✗	✗	✗
RealEstate10K [221]	~10,000	 	Real	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗
DL3DV-10K [97]	10,510	 	Real	✓	✗	✗	✗	✗	✓	✗	✗	✓	✗
Datasets for other downstream tasks													
SPair-71k [116]	— (70,958 pairs)		Real	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
SIMPLER [93]	8		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
CALVIN [113]	34		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
MetaWorld [196]	50		Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
RoboTwin 2.0 [20]	50	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
HOI4D [102]	57		Real	✗	✓	✓	✗	✓	✗	✗	✓	✗	✓
ManiSkill 3 [159]	58	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
DROID [81]	86	 	Real	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
LIBERO [99]	130	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓
MVRefer [179]	1,513	 	Real	✗	✗	✗	✗	✗	✗	✗	✓	✗	✗
Cityscapes [29]	5,000		Real	✗	✓	✗	✗	✗	✗	✗	✓	✓	✗
NAVSIM [34]	115,000		Real	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Ego-Exo4D [54]	131	 	Real	✓	✗	✗	✗	✓	✗	✗	✓	✓	✓
HolIHQ [214]	3,000	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗
Scene-Decoupled-Video-dataset [70]	3,400	 	Synthetic	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗

and point predictions. Efficiency targets retain the same geometric outputs while changing view count, resolution, numerical precision, token budget, architecture, or hardware; runtime, memory, computation, and geometric accuracy are then reported for a single forward-pass setting. Robustness targets instead vary the type and severity of visual degradation, environmental interference, occlusion, or transient distractors. Because the output tasks overlap, these conditions must be matched before accuracy, latency, memory, efficiency, or robustness gains are compared.

Temporal targets cover streaming, long-sequence, and dynamic reconstruction. Streaming and long-sequence evaluation combines trajectory drift and retained geometry with sequence and window length, cache or state size, chunk overlap, alignment, loop closure, revisit conditions, and whether observations are processed offline or causally. Dynamic targets further separate camera motion, static structure, moving regions, and dense 3D motion through trajectory, mask, flow, and reconstruction metrics. Reports must pair each temporal target with the geometry retained over the evaluated sequence.

Table 5. Evaluation for geometric state strengthening. Each group lists its evaluation focus, representative datasets, key metrics, and comparison conditions. Common geometric measures are summarized above the groups.

Evaluation focus	Representative datasets	Key metrics	Key comparison conditions
Common geometry metrics, selected by task:			
Pose: relative rotation / translation accuracy (RRA / RTA), pose area under the curve (AUC).			
Depth: absolute relative error (AbsRel), root mean squared error (RMSE), threshold accuracy at 1.25 (δ_1).			
Surface: Chamfer distance (CD), accuracy, completeness.			
Diverse-input 3D reconstruction			
Geometry with diverse inputs	Pose: CO3Dv2 [133] Depth: NYU-v2 [146] Surface: DTU [73] Multimodal: ViViD++ [86] Event: MVSEC [223] Panoramic: Replica360-2K [136]	Pose / depth / surface metrics for the supplied inputs	Input modality / sparsity / noise; camera model / calibration; view count / metric scale; backbone adaptation
Efficient 3D reconstruction			
Geometry quality and computation	Pose: CO3Dv2 [133] Geometry: ETH3D [141]; ScanNet [30]	Geometry metrics; FPS / latency; peak GPU memory; FLOPs / bits per weight	Views / resolution / batch size; precision / token budget; architecture / hardware; timing scope
Robust 3D reconstruction			
Geometry under adverse conditions or outlier views	Adverse conditions: Dark3R [58]; FIORD [57] Outlier views: ETH3D [141]	Depth: AbsRel / RMSE _{log} / δ_1 ; pose: RRA / RTA / AUC; rotation / translation-direction error	Environment / degradation severity; distractor ratio / view overlap; paired clean and degraded inputs; adaptation
Streaming and long-sequence 3D reconstruction			
Trajectory drift, retained geometry, and resource use	Trajectory: TUM RGB-D [149]; KITTI Odometry [52] Long sequences: TartanAir [177]; Waymo [152]	Geometry metrics; ATE / RPE; translation / rotation drift; FPS / peak GPU memory	Offline / causal processing; sequence / window length; chunk overlap / stride; memory budget / warm-up; alignment / loop closure / revisits
Dynamic 3D reconstruction			
Dynamic geometry, motion, and segmentation	Camera / depth: Bonn [119] Motion: PointOdyssey [216]; Kubric [55] Masks / views: DAVIS [128]; DyCheck [47]	Geometry metrics / ATE; 3D endpoint error; mask IoU / boundary F-score	Static-dynamic split; frame span / camera motion; mask availability; scale / temporal alignment

ATE: absolute trajectory error; RPE: relative pose error; RMSE_{log}: RMSE of log-depth; FPS: frames per second; FLOPs: floating-point operations; IoU: intersection over union.

5.3 Evaluation Targets for Downstream State Reuse

Current downstream targets measure the resulting application capability rather than the contribution of the geometric-state information used by the application. Because the 5 applications produce different outputs, we examine their targets directly in the order of NVS, SLAM, semantic 3D scene understanding, world models, and embodied applications. Table 6 links evaluation tasks to representative datasets, key metrics, and comparison conditions within each application group.

Table 6. Evaluation for downstream geometric-state reuse. Rows link evaluation tasks to representative datasets, key metrics, and comparison conditions within each application group.

Evaluation task	Representative datasets	Key metrics	Key comparison conditions
NVS			
Static-scene NVS	RealEstate10K [221]; DL3DV-10K [97] CO3Dv2 [133]; Mip-NeRF 360 [6]	PSNR / SSIM / LPIPS	Input / target views; cameras / resolution; scene-specific optimization
Dynamic-scene NVS	DAVIS [128]; DyCheck [47]; HyperNeRF [126]	PSNR / SSIM / LPIPS	View / temporal split; masks / dynamic content; scene-specific optimization
SLAM			
Tracking and mapping	TUM RGB-D [149]; EuRoC [9]	Trajectory: ATE / RPE; map: accuracy / completeness	Trajectory / alignment; loop closure / map retention
Visual odometry	KITTI [52]; TUM-VI [142]	ATE / RPE; translation / rotation drift	Trajectory length; scale / pose alignment
Relocalization	Cambridge Landmarks [79]; LaMAR [138]	Translation / rotation error; localization recall	Query / reference split; map / pose thresholds
Place recognition	NCLT [16]; Oxford RobotCar [108]	Recall@K	Query / retrieval database; matching threshold / K
Semantic 3D scene understanding			
3D semantic prediction and mapping	ScanNet200 [137]; ARKitScenes [7]	Prediction: mIoU / AP	Class vocabulary; 2D–3D label mapping
	SPair-71k [116]	Correspondence: PCK	View split; keypoint threshold
Language-based 3D scene reasoning	MVRefer [179]	Grounding accuracy at IoU 0.25	Language queries; view split; grounding threshold
Geometry-aware world models			
Video-depth prediction	Cityscapes [29]; TartanAir [177]	AbsRel / RMSE	Prediction horizon; camera conditioning
Video generation	HolHQ [214]; Scene-Decoupled-Video-dataset [70]	Visual: PSNR / SSIM / LPIPS / FVD; geometry: reprojection error / PCK	Camera / interaction conditions; sampling budget; teacher forcing
Action prediction / rollout selection	DROID [81]; LIBERO [99]	Task success rate	Action conditioning; horizon / rollout budget; selection policy
Embodied action and planning			
Manipulation	ManiSkill 3 [159]; RoboTwin 2.0 [20]; DROID [81]	Task success rate	Task suite / demonstrations; cameras / control horizon
Long-horizon policies / generalization	LIBERO [99]; CALVIN [113]; SIMPLER [93]	Task success rate	Task sequence / test conditions; policy initialization; encoder adaptation
Driving	NAVSIM [34]	PDM score	Scenario split; camera setup / control horizon

PSNR: peak signal-to-noise ratio; SSIM: structural similarity index measure; LPIPS: learned perceptual image patch similarity; FVD: Fréchet video distance; mIoU: mean intersection over union; AP: average precision; PCK: percentage of correct keypoints. Geometric metric abbreviations follow Table 5.

NVS targets use held-out views to measure appearance and, where geometry is available, surface fidelity. Input and target views, camera information, resolution, scene-specific optimization, masks, and dynamic content must be aligned before rendering or geometry scores are compared. High rendering quality alone does not establish accurate surfaces.

SLAM targets combine trajectory and map evaluation with retrieval and metric-pose evaluation for relocalization. Trajectory segments, alignment rules, loop-closure settings, map retention, retrieval databases, and decision thresholds must be matched; low local trajectory drift can otherwise coexist with global map loss.

Semantic-understanding targets evaluate objects, regions, occupancy, correspondence, cross-epoch change, language grounding, or spatial reasoning. Because geometry is assessed through task-specific annotations, comparisons must align class vocabularies, 2D–3D label mappings, view and temporal splits, registrations, query sets, and matching thresholds.

World-model targets extend observed geometry to future observations or actions. They assess geometry forecasting, future-view or video generation, and action-conditioned rollouts using geometric consistency, visual fidelity, and, where applicable, task success. Prediction horizon, conditioning, sample budget, and rollout-selection rule must be specified because a visually plausible single-step future does not establish persistent geometry or useful long-horizon state transitions.

Embodied targets report task or manipulation success rates, path efficiency, collision rates, driving comfort, or trajectory quality. Geometry is evaluated indirectly through action outcomes, so the task suite, demonstrations, camera setup, control horizon, policy initialization, and encoder adaptation must be fixed before gains are attributed to the geometric-state information supplied to the application.

Across these applications, downstream scores—such as segmentation accuracy, video quality, manipulation success rate, and trajectory quality—characterize the resulting capability. The next subsection summarizes the observed limitations in dataset coverage and target comparability.

5.4 Limitations of Current Datasets and Evaluation Targets

The reviewed tasks do not have equal evaluation coverage. Static RGB geometry and NVS have comparatively mature datasets and targets, whereas long streams, dynamic scenes, specialized sensors, semantic persistence, embodied interaction, and world-model prediction rely on smaller or task-specific collections. Few datasets contain the geometric, temporal, semantic, and action annotations needed to evaluate several capabilities on the same scenes.

Evaluation targets also vary within a dataset. Input count, image resolution, temporal window, train–test split, scale alignment, model adaptation, and hardware are not standardized across papers. Moreover, current targets rarely test what geometric information $\mathcal{Z}_{\text{geo}}$ retains, and cameras, depth maps, point maps, and tracks in $\mathcal{R}_{\text{geo}}$ are often scored separately without checking whether they describe consistent geometry. Dataset coverage is therefore uneven, and results obtained under different target settings are not directly comparable.

6 Open Challenges

The preceding sections show that progress is uneven. Feed-forward reconstruction is well established for bounded, predominantly static image sets, and later methods address diverse inputs, resource limits, difficult sensing, long streams, and scene motion. Reuse is also well developed for rendering and geometric mapping, but remains less established for semantic, geometry-aware world-model, and embodied applications. The reviewed evidence exposes 3 field-level problems concerning the reliability of geometric state, its adaptation to different applications, and dataset and benchmark design for identifying the contribution of geometric state:

- How can geometric state remain reliable and persistent as measurements, sequence length, and scene motion change?
- How can geometric representations adapt to the different information required by downstream applications?
- How should datasets and benchmarks be designed to determine whether application gains result from changes in or reuse of geometric state?

The 3 subsections below discuss these problems in turn, summarizing current evidence, the remaining gap, and promising research directions.

6.1 Reliable and Persistent Geometric State

Recent work has strengthened feed-forward reconstruction in several aspects. Diverse-input models accept camera parameters, depth, partial geometry, LiDAR, or non-standard visual inputs; robust models address degraded and ambiguous visual evidence; efficient models reduce the cost of a forward pass; streaming and long-sequence models transfer and align state across windows; and dynamic models represent scene motion. These advances extend $\mathcal{Z}_{\text{geo}}$ and $\mathcal{R}_{\text{geo}}$ beyond clean, fixed RGB image sets, but they do not yet provide a single state that remains reliable as sensing conditions, scene extent, and time change together.

The first problem is that a supplied measurement is not guaranteed to remain in the final geometry. In many diverse-input models, known depth, metric scale, calibration, or LiDAR samples are encoded into $\mathcal{Z}_{\text{geo}}$ and decoded through learned heads. The resulting structured geometric outputs can improve without showing which measured values were preserved, which regions were completed by the model, or which predictions are unreliable. MapAnything [78] and LiDAR-VGGT [171] make geometric constraints more explicit, but they do not establish a common test for measurement preservation and uncertainty.

The second problem is that no reviewed system simultaneously handles bounded computation and memory, a stable global coordinate frame, reliable recognition of revisited places, and persistent identities for moving objects as a sequence grows. StreamVGGT [224] and FrameVGGT [186] explore cached or bounded short-term context, Anchor3R [158] introduces transient anchors for long-horizon alignment, and PAGE-4D [220] and DynamicVGGT [63] model scene motion. These components remain separate, however, and have not been unified with the diverse sensing settings reviewed in Section 3.1. A short-term cache may forget global anchors, a compact state may discard observations needed later, and a static map may absorb moving objects into inconsistent geometry. The open question is how a single system can retain measured geometry, maintain global consistency, and separate persistent structure from change over long streams.

A useful direction is to retain measurement-linked geometry with explicit uncertainty and provenance, together with short-term information for current reconstruction, long-term anchors for scale and revisits, and object records for identity and motion. These components should remain geometrically aligned while being updated at different rates.

6.2 Adaptive Geometric Representations for Downstream Applications

The 5 application groups need different parts of geometric state. NVS uses surfaces, viewpoints, and appearance; SLAM uses a stable coordinate frame, trajectories, and long-term anchors; semantic applications attach objects and language to 3D locations; world models use state that changes with time and action; and embodied applications need free space, contact geometry, and reliability for action. Existing systems show that structured geometric outputs, multi-view geometry latent features, or their combination can support each individual requirement.

The problem is that these choices are made separately for each application. Some methods use one decoded output, whereas others use multi-view geometry latent features or distilled features, and their comparisons use different architectures and targets. Passing every feature and output is costly, while discarding information can remove what another application needs. There is still no common geometric state that can expose the information needed by different applications without rebuilding a separate representation for each application.

A useful direction is an adaptable state with stable geometric anchors and task-specific access: surfaces and cameras for rendering, camera trajectories and persistent landmarks for mapping, object-linked geometry for semantics, local constraints for action, and time-varying entities for prediction. Matched comparisons of $\mathcal{R}_{\text{geo}}$, $\mathcal{Z}_{\text{geo}}$, and their combination should test which parts are needed for each capability under the same data, supervision, and model capacity. The goal is a common state that applications can select and adapt to their needs.

6.3 Dataset Design for Benchmarking Geometric State

The central evaluation problem is whether an application gain actually comes from geometric-state reuse. A higher rendering, mapping, semantic, prediction, or action score does not answer this question when a method also changes the application architecture, adds supervision, increases model capacity, or introduces task-specific adaptation. Current results therefore establish that the complete application improved, but often cannot isolate the contribution of $\mathcal{Z}_{\text{geo}}$ or $\mathcal{R}_{\text{geo}}$.

A direct test requires a controlled comparison within the same application. The application model, training data, supervision, and training budget remain unchanged, while its input is replaced in turn by $\mathcal{R}_{\text{geo}}$, $\mathcal{Z}_{\text{geo}}$, their combination, and a baseline without VGGT geometric state. If an application improves only when geometric-state information is provided, the performance difference can be attributed to that information more directly. This comparison also reveals whether the application needs structured geometric outputs, multi-view geometry latent features, or their combination.

A benchmark for this purpose must make the controlled comparison reproducible. It needs a fixed data split, a fixed application model and training recipe, and evaluation targets for both geometry and the application outcome. Each geometric-state input is then tested under the same evaluation setup, and the benchmark reports both the change in geometric quality and the corresponding change in application performance. This design would show whether improving or using the geometric state leads to a measurable application gain, rather than merely showing that two different complete systems obtain different scores.

7 Conclusion

This survey organizes the VGGT [167] lineage around one geometric state with two components: the multi-view geometry latent features $\mathcal{Z}_{\text{geo}}$ and the structured geometric outputs $\mathcal{R}_{\text{geo}}$. State-strengthening work improves how this state is formed from diverse inputs, computed efficiently, made robust, retained over streams, and extended to dynamic 3D reconstruction. State-reuse work uses structured geometric outputs, multi-view geometry latent features, or their combination for NVS, SLAM, semantic 3D scene understanding, geometry-aware world models, and embodied action and planning.

The dataset and benchmark analysis reveals uneven evaluation coverage: static reconstruction and rendering are well supported, whereas long-stream, dynamic, and task-oriented settings rely more heavily on specialized resources. Open challenges include maintaining reliable and persistent geometric state, adapting representations to different applications, and designing datasets and benchmarks that isolate whether application gains arise from changes in or reuse of geometric state. Addressing these challenges can extend VGGT [167]-based geometric state from isolated reconstruction outputs to broader applications.

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